Exercise 3 – Classification I

Introduction to Machine Learning

Exercise 1: Logistic regression – softmax link and decision boundary

The whole exercise is only for lecture group A!

Learning goals

1. Solve "show equivalence"-type questions
2. Derive the decision boundary of the logistic classifier

Binary logistic regression is a special case of multiclass logistic, or softmax, regression. The softmax function is the multiclass analogue to the logistic function, transforming scores $\boldsymbol{\theta}^\top \mathbf{x}$ to values in the range [0, 1] that sum to one. The softmax function is defined as:

$$\pi_k(\mathbf{x}|\boldsymbol{\theta}) = \frac{\exp(\boldsymbol{\theta}_k^\top \mathbf{x})}{\sum_{j=1}^{g} \exp(\boldsymbol{\theta}_j^\top \mathbf{x})}, k \in \{1, ..., g\}$$

a) Show that logistic and softmax regression are equivalent for $g = 2$.

b) Recall that in the binary case we predict $\hat{y} = 1$ iff $\pi(\mathbf{x}) \geq \alpha$ for a threshold $\alpha \in (0, 1)$. Show that the resulting decision boundary is a (linear) hyperplane.

Hint

Derive the value of $\boldsymbol{\theta}^\top \mathbf{x}$ (depending on α) starting from which you predict $\hat{y} = 1$ rather than $\hat{y} = 0$.

Exercise 2: Hyperplanes

Learning goals

1. Understand that hyperplanes bisect the space with a linear boundary
2. Get a feeling for coefficients in hyperplane equations
3. See how the classification threshold α shifts the logistic decision boundary

Linear classifiers like logistic regression learn a decision boundary that takes the form of a (linear) hyperplane. Hyperplanes are defined by equations $\boldsymbol{\theta}^\top \mathbf{x} = b$ with coefficients $\boldsymbol{\theta}$ and a scalar $b \in \mathbb{R}$.

a) In order to see that such expressions actually describe hyperplanes, consider $\boldsymbol{\theta}^\top \mathbf{x} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$. Sketch the hyperplanes given by the following coefficients and explain the difference between the parameterizations:

- $\theta_0 = 0, \theta_1 = \theta_2 = 1$
- $\theta_0 = 1, \theta_1 = \theta_2 = 1$
- $\theta_0 = 0, \theta_1 = 1, \theta_2 = 2$

b) Beyond its orientation, a linear classifier also has an *offset* that determines where exactly the boundary sits. Recall that a logistic regression model estimates $\pi(\mathbf{x}) = \frac{1}{1+\exp(-\boldsymbol{\theta}^\top \mathbf{x})}$ and predicts $\hat{y} = 1$ iff $\pi(\mathbf{x}) \geq \alpha$ for a threshold $\alpha \in (0, 1)$. Its decision boundary is therefore the hyperplane

$$\boldsymbol{\theta}^\top \mathbf{x} = -\log\left(\frac{1}{\alpha} - 1\right) =: b(\alpha).$$

Consider the fixed model $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)^\top = (0, 1, 1)^\top$. Compute the offset $b(\alpha)$ for $\alpha \in \{0.25, 0.5, 0.75\}$, sketch the three resulting decision boundaries in a single (x_1, x_2) plot, and describe how the boundary – and the region predicted as class 1 – changes as α grows. Does the orientation of the boundary change?

Exercise 3: Decision Boundaries & Thresholds in Logistic Regression

Learning goals

1. Understand that logistic regression finds a linear decision boundary
2. Get a feeling for how the parameters $\boldsymbol{\theta}$ shape predicted probabilities and model confidence
3. Understand the role of the decision threshold α

Recall from Exercise 2 that a logistic regression model estimates the probability $p(y = 1|\mathbf{x}, \boldsymbol{\theta}) = \pi(\mathbf{x}|\boldsymbol{\theta})$ and assigns $\hat{y} = 1$ iff $\pi(\mathbf{x}|\boldsymbol{\theta}) \geq \alpha$ for a threshold $\alpha \in (0, 1)$. Whereas Exercise 2 b) varied the threshold for a fixed model, we now study how the parameters $\boldsymbol{\theta}$ themselves shape the predicted probabilities.

a) Below you see the logistic function for a binary classification problem with two input features for different values $\boldsymbol{\theta}^\top = (\theta_1, \theta_2)^\top$ (plots 1-3) as well as α (plot 4). What can you deduce for the values of θ_1, θ_2, and α? What are the implications for classification in the different scenarios? In particular, which of the plots (1)–(3) shows the most confident model?

b) In Exercise 2 b) we simply read off the offset $b(0.5) = 0$ for the common threshold $\alpha = 0.5$. Derive this special case directly, starting from $\pi(\mathbf{x}) = 0.5$.

c) Explain when it might be sensible to set α to 0.5.

Exercise 4: Thresholds and predictions in practice

The whole exercise is only for lecture group B!

Learning goals

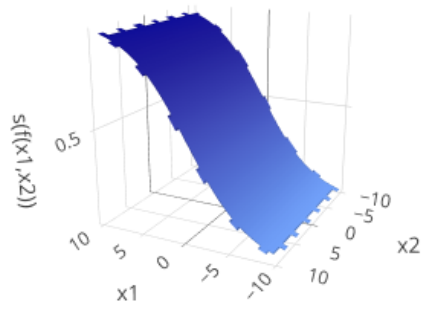
1. Compute predictions of a logistic model by hand
2. Understand how the threshold α trades off the two error types

Consider a fitted binary logistic regression model with parameter vector $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)^\top = (-1, 2, 1)^\top$, so that

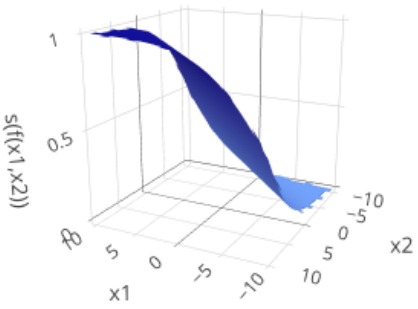
$$\pi(\mathbf{x}) = \frac{1}{1 + \exp\left(-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)\right)}.$$

Recall that we predict $\hat{y} = 1$ iff $\pi(\mathbf{x}) \geq \alpha$.

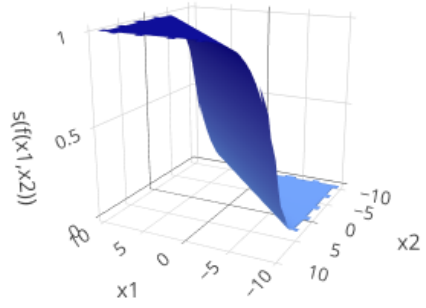
Plot (1)



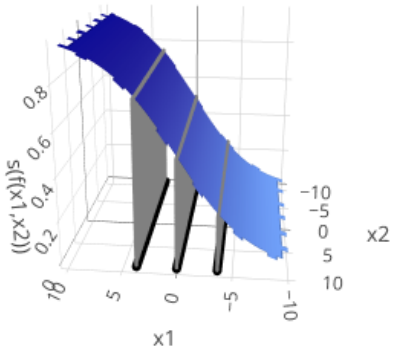
Plot (2)



Plot (3)



Plot (4)



a) Compute the predicted probability $\pi(\mathbf{x})$ for the following observations and classify them using the default threshold $\alpha = 0.5$:

- $\mathbf{x}^{(1)} = (x_1, x_2)^\top = (1, 1)^\top$
- $\mathbf{x}^{(2)} = (1, -1)^\top$
- $\mathbf{x}^{(3)} = (0, 0)^\top$

b) For observation $\mathbf{x}^{(1)}$, from which threshold α onwards would the model predict $\hat{y} = 0$ instead?

c) The model is used in a hospital to decide whether to start an expensive and invasive therapy ($\hat{y} = 1$). Would you rather choose a high or a low threshold α? Which type of error (false positive / false negative) does your choice avoid, and at what cost?